BEHIND SENSATIONS

PYQ • 2021 - 2026

1. 2026 **1 MARK**

Analyse the process of taste detection and choose the correct sequence of steps from the options given :

Steps : (i) Substances dissolve in saliva. (ii) Nerves carry impulses to brain.

(iii) Chemoreceptors generate impulses. (iv) Substances reach the minute pores in the papilla.

(a) (iii), (ii), (i), (iv) (b) (ii), (i), (iii), (iv)

(c) (i), (iv), (iii), (ii) (d) (i), (iv), (ii), (iii)

2. 2026 **2 MARKS**

Analyse the figure given and answer the following questions.

(a) Identify the figure related to clear vision in dim light.

(b) Which muscular activities take place here ?



3. 2026 **4 MARKS**

(A) Redraw the diagram. Label it by identifying the parts indicated in the statements given.

(a) The part that allows light to enter the eye.

(b) The part that adjusts the curvature of the lens.

(c) The part where the cone cells are abundant.

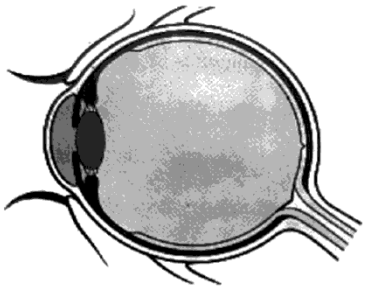
OR

(B) Observe the illustration and answer the following questions.

(a) Identify the parts labelled as 'X' and 'Y'.

(b) What will be the effect of the rotational movement of head in 'X' ?

(c) Write the processes in 'Y' that leads to hearing.



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4. 2025 **1 MARK**

Which of the following activity is correct while viewing nearby objects ?

(i) The ciliary muscles contract. (ii) The curvature of the lens decreases.

(iii) The ligaments stretch. (iv) The focal length decreases.

(a) (i) and (ii) (b) (i), (ii) and (iii) (c) (i) and (iv) (d) (ii), (iii) and (iv)

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5. 2025 **3 MARKS**

Analyse the table showing defects or diseases and their causes, then answer the questions.

Xerophthalmia : ... (i) ...

... (ii) ... : Defects in cone cells

... (iii) ... : Reabsorption of aqueous humor does not occur

Night blindness : ... (iv) ...

(a) Identify and write (i), (ii), (iii) and (iv).

(b) What is the treatment for the disease indicated by (iii) ?

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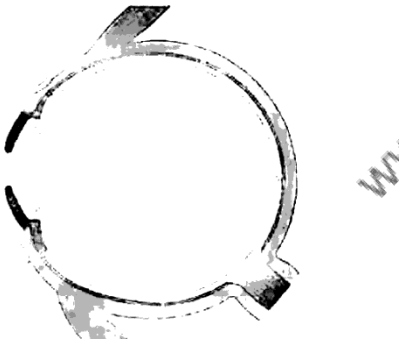
6. 2025 **4 MARKS**

Redraw the diagram, identify and label the parts with their names.

(a) The part that allows light rays to enter the eye.

(b) The muscles that adjust the curvature of the lens.

(c) The part that adjusts its size according to the intensity of light.



7. 2024 **1 MARK**

Choose the correct combination from the following :

(i) Planaria : Eye spot : Taste (ii) Shark : Jacobson's organ : Smell

(iii) Snake : Lateral line : Body Balance (iv) House fly : Ommatidia : Light

(a) (i), (ii) correct (b) (ii) correct (c) (ii), (iii) correct (d) (iv) correct

8. 2024 **2 MARKS**

Read the following statement and answer the question.

'The middle ear is separated from the external ear by the tympanum, air cannot pass through it'.

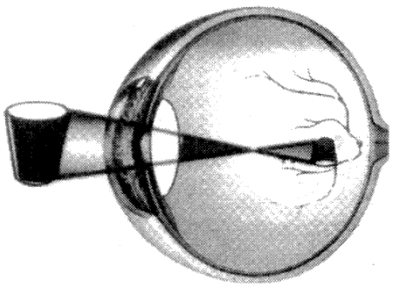
So how is the air pressure on either side of the tympanum regulated ?

9. 2024 **2 MARKS**

Observe the illustration related to the sense of vision and answer the questions.

(a) Why is this type of image formed in the retina ?

(b) Even though such an image is formed, how can we see the object in its true form ?



10. 2024 **3 MARKS**

Prepare a flow chart related to the sense of taste by selecting the appropriate information given in the box.

Stimulate the olfactory receptors ; Substances responsible for taste dissolve in saliva ; Impulses reach the cerebrum ; Aromatic particles enter the nostrils ; Generate impulses ; Enter in the taste bud ; Experience taste ; Stimulate the chemoreceptors ; Aromatic particles dissolve in the mucus ; Impulses reach the cerebellum

11. 2023 **1 MARK**

Identify the word pair relation and fill in the blank.

Planaria : Eye spot

House fly : _____

12. 2023 **3 MARKS**

Give reason for the following.

(a) There is no vision in the Blind spot.

(b) Persons with colour blindness are not selected for jobs like that of a driver, pilot etc.

(c) The deficiency of Vitamin A results in Night blindness.

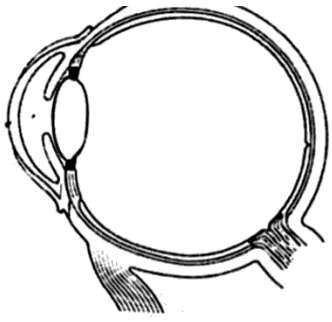
13. 2023 **4 MARKS**

Redraw the diagram. Name and label the parts indicated below. (For drawing 1 score)

(a) The transparent anterior part of the sclera.

(b) The point of maximum visual clarity.

(c) Transmits impulses to visual centre of the brain.



14. 2022 **1 MARK**

Identify the given photoreceptor cell.



15. 2022 **1 MARK**

Housefly : Ommatidia

Planaria : _____

16. 2022 **3 MARKS**

Analyse the given table and arrange Columns B and C according to Column A.

A : Cornea, Pupil, Yellow spot

B : The part of the retina where plenty of photoreceptors are present ; The projected transparent anterior part of the sclera ; The part of the retina from where the optic nerve begins ; The aperture seen at the centre of the iris

C : Transmits impulses from photoreceptors to the visual centre in the brain ; Regulates the amount of light falling on the eyes ; Refracts light rays to focus on the retina ; Point of maximum visual clarity

17. 2022 **4 MARKS**

The feature of a fluid in the eye is given below. Analyse it and answer the following questions.

'It is formed from blood, and is reabsorbed into blood'

(a) Name the fluid.

(b) What is its function ?

(c) Name the eye disease associated with this fluid.

(d) How does this disease can be rectified ?

18. 2021 **1 MARK**

Which of the following statement is true about yellow spot ?

(a) There is no vision at this point. (b) Photoreceptors are absent here.

(c) The optic nerve begins from this point. (d) This is the point of maximum visual clarity.

19. 2021 **2 MARKS**

Correct mistakes if any in the underlined part of given statements.

(a) Substances responsible for taste should dissolve in mucus to stimulate the chemoreceptors.

(b) Cochlea contains specialised sensory hair cells concerned with hearing.

(c) Colour blindness is a condition due to the defect of the cone cells.

(d) The cluster of photoreceptors of a housefly is eye spot.

20. 2021 **2 MARKS**

Observe the picture and answer the questions given below.

(a) Identify the photoreceptor.

(b) Name the pigment present in this receptor.

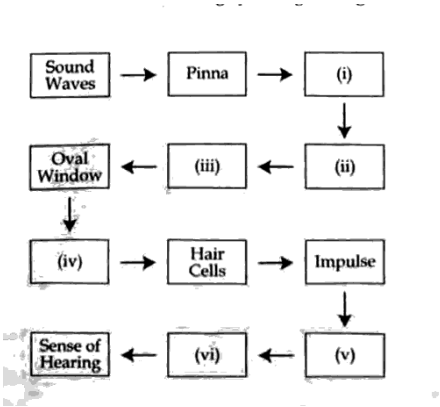
(c) Write the function of this pigment.

(d) Name the eye defect caused due to the deficiency of this pigment.



21. 2021 **3 MARKS**

Complete the flow chart related to hearing by adding missing terms.



22. 2021 **3 MARKS**

Processes in the sense of sight is given below. Arrange them in sequential order.

• Sense of sight

• Lens focus the light rays into the retina.

• Cornea refracts light into the eye.

• Impulse transmit through optic nerve to the cerebrum.

• Photosensory cells on the retina stimulated.

• Light enter through the pupil falls on the lens.

23. 2021 **4 MARKS**

Redraw the figure. Identify, name and label the following parts.

(a) The projected transparent anterior part of sclera.

(b) The aperture seen at the centre of the iris.

(c) The nerve that transmit impulses to the brain.

